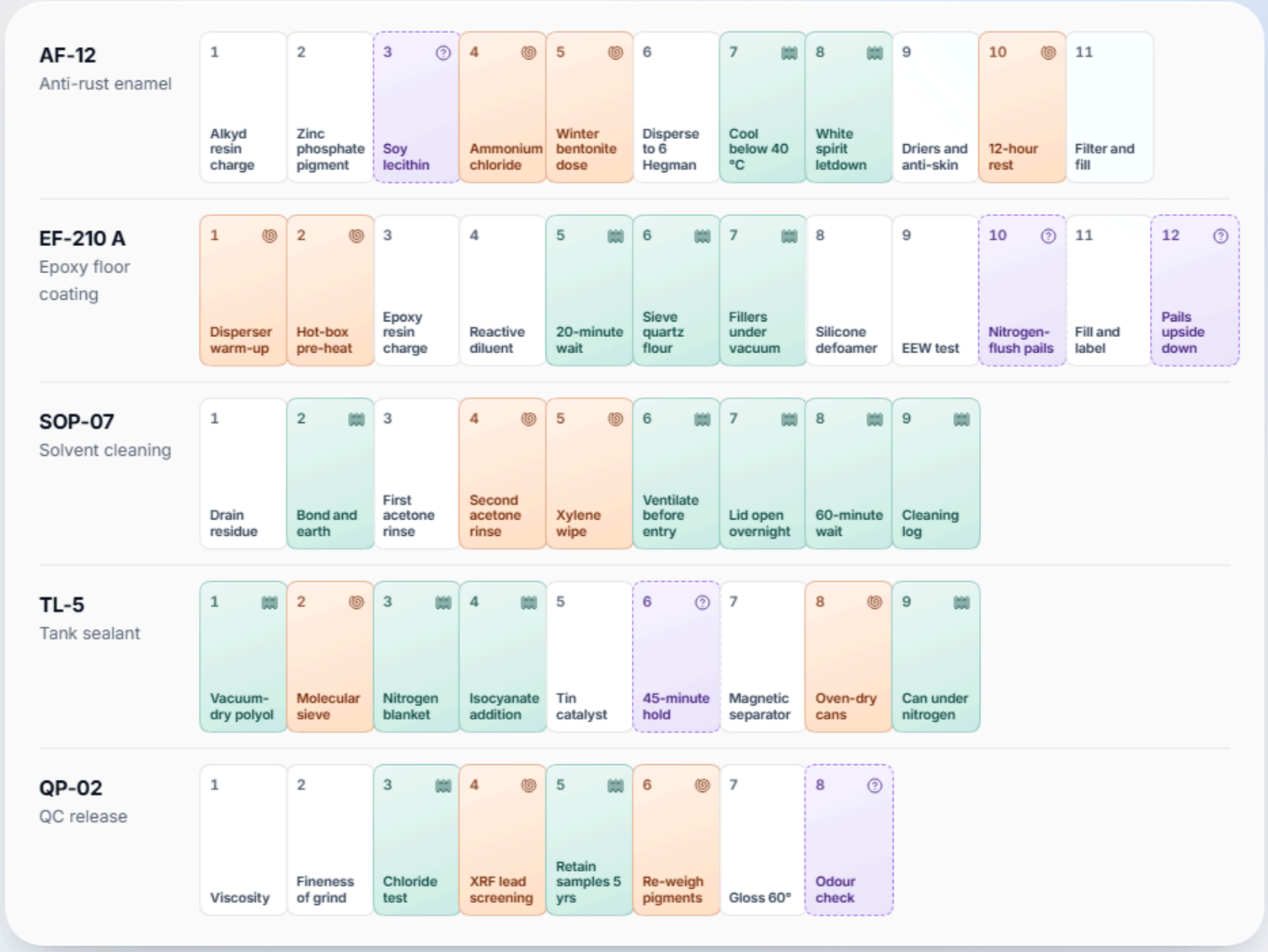
24 Cr CROMO

Every factory runs on reasons nobody remembers.

CROMO reads a plant's own archive and recovers why each process step exists: it finds the waste and protects the safeguards.



€305,298 /yr
in 11 fossil steps whose reason is gone

16 fences
protected

5 unknown
sent to a person

Fossil Fence Unknown

Google Gemma 4 on the AMD Token Factory · Evolus

1975 · PRIMO LEVI, THE PERIODIC TABLE, "CHROMIUM"

A batch of anti-rust paint keeps turning to jelly.

A young chemist traces it to one bad pigment delivery and fixes it by adding ammonium chloride. **Twenty years later** the bad pigment is long gone, and the factory still adds ammonium chloride to every batch. Nobody can say why.

The opposite also happens. Chesterton (1929): *don't tear down a fence until you know why it was put up.* Some steps that look pointless are load-bearing.

Cromo is Italian for chromium.

THE PROBLEM

Recipes are full of steps whose reasons retired with the people who wrote them.

Fossils cost money every batch: material, labour, energy, tank time. Nobody dares remove them.

Fences look like waste but prevent fires, recalls and claims. Remove one and the old accident comes back.

And AI optimisers now propose "remove this, shorten that", without knowing which is which.

Users: process engineers · quality / EHS managers · plant managers. A human approves every change.

WHY IT MATTERS · A REAL CASE

Flixborough, UK, 1974: one plant change, made without a full assessment.

A cracked reactor was taken out and a temporary bypass pipe joined its neighbours so production could go on: no drawing, limited calculations, no pressure test. On 1 June 1974 the bypass ruptured, the escaping cyclohexane exploded, and **28 workers were killed**.

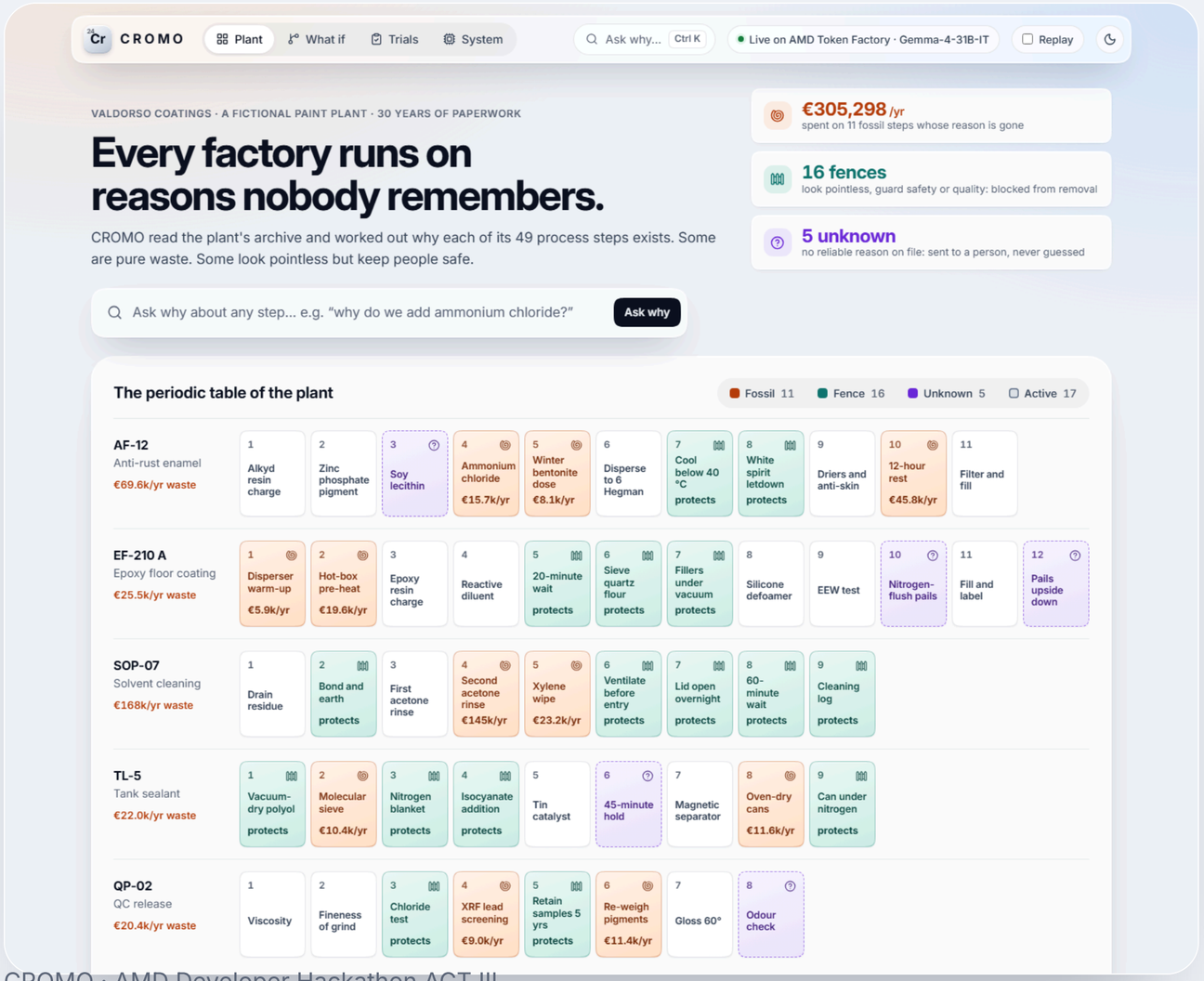
It became a turning point for process safety and for what is now called **management of change**. Today OSHA's Process Safety Management standard requires the technical basis and the safety impact of a change to be addressed **before** it is made.

CROMO makes no claim about any past incident. It helps recover and check the reasons behind a step before anyone changes it.

Sources: UK Health and Safety Executive, "Flixborough (Nypro UK) Explosion 1st June 1974" case study (citing the 1975 Court of Inquiry report) · IChemE Safety and Loss Prevention SIG, "50 Years on from Flixborough" · US OSHA, 29 CFR 1910.119(l), Management of change.

THE PRODUCT · COMPUTED ON AMD BY GEMMA-4-31B-IT

The periodic table of a plant.



€305,298/yr

in **11 fossil steps** of a synthetic mid-sized coatings plant

16 fences

protected from removal

5 unknown

sent to a person, never guessed

synthetic data · fictional plant

THE WHY-TRAIL · THE LEVI CASE, RECREATED IN OUR ARCHIVE · AS CITED BY THE AMD RUN

It digs, it does not chat.

1995 Deviation report · In 1995, livering was suspected to be caused by basic lead chromate pigment lots...

1995 Lab test report · In 1995, ammonium chloride was found to neutralize excess alkali in the pigment...

1995 Change request · In 1995, ammonium chloride was added to AF-12 to prevent livering caused by high...

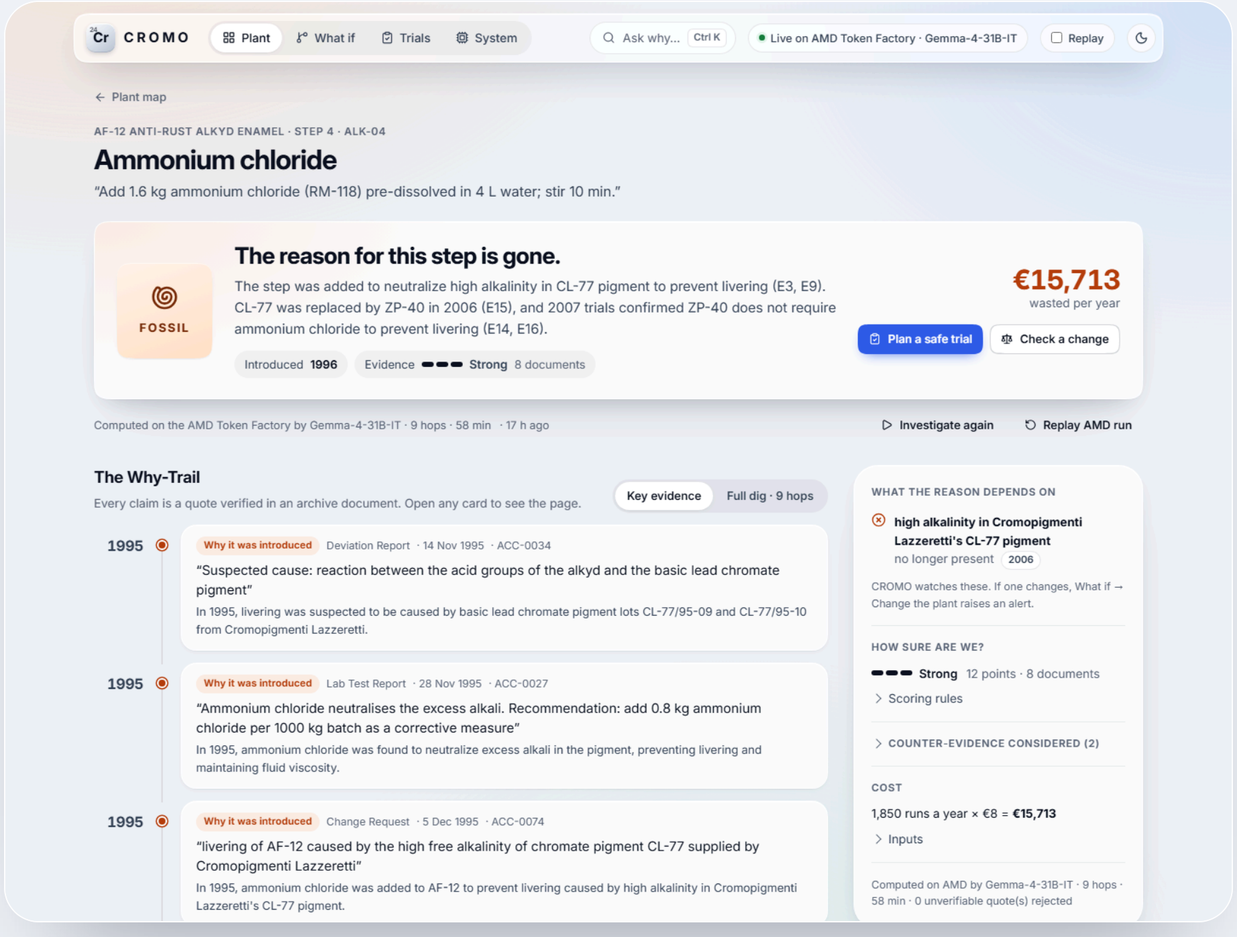
1996 Recipe · Earliest archived recipe revision containing the step is dated 1996-01-22.

1996 Change request · In 1996, a chloride limit of 400 ppm was introduced for AF-12 because ammonium chloride...

2006 Compliance memo · In 2006, the lead chromate pigment CL-77 from Cromopigmenti Lazzeretti was phased out...

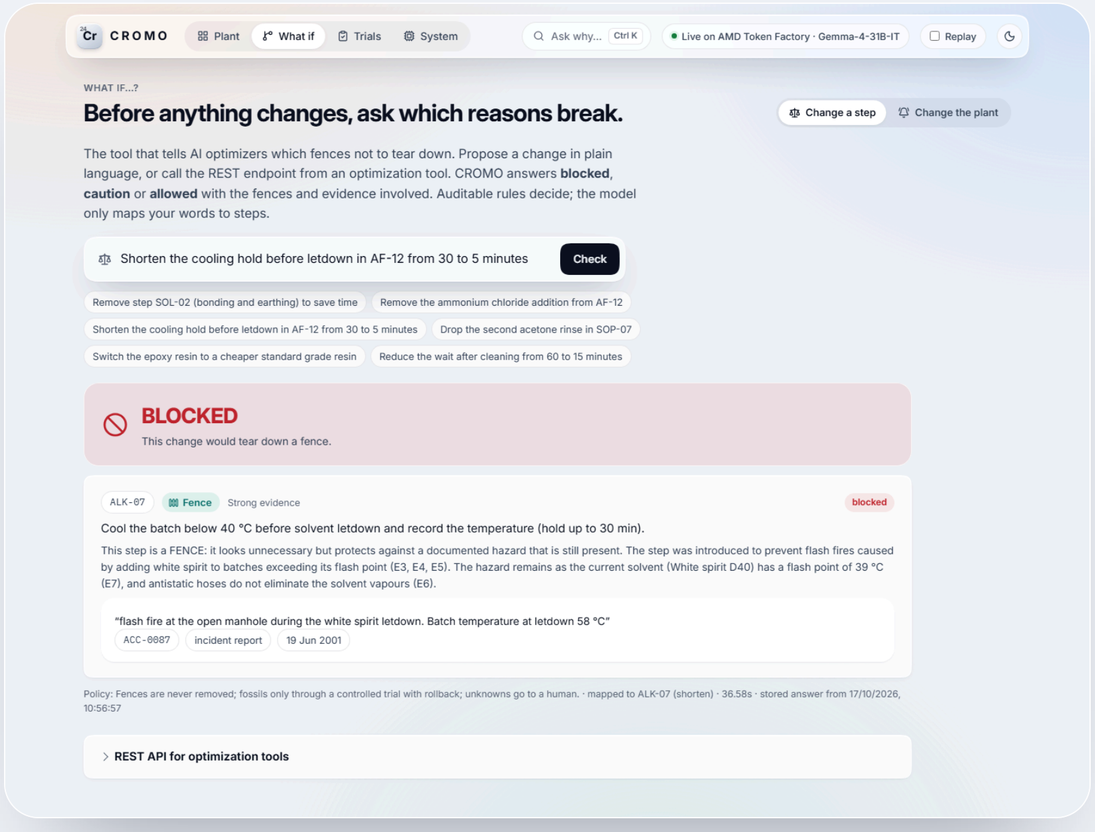
2020 Lab test report · In 2020, ammonium chloride was linked to under-film corrosion complaints on AF-12...

FOSSIL Strong evidence · €15,713/yr · 9 hops



FENCES AND GUARDRAILS

Models read the evidence. Rules decide what may be removed.



Every claim cites a quote **verified verbatim** in the source; uncited claims are dropped.

A fossil needs proof that its reason ended and no open safety evidence; otherwise it goes to a person.

What if...? Change a step: blocked · caution · allowed with trial (also a REST API for optimisers). Change the plant: tripwires show which reasons break.

HOW IT WORKS · EVERY MODEL CALL ON THE AMD TOKEN FACTORY (SHARED ENDPOINTS ON AMD GPUS)

Read the archive. Follow the paper trail. Decide with rules.

Role	Model	Runs on
Vision: scans, faxes, handwritten Italian logbooks	**Gemma 4 31B-IT (Google DeepMind)**	AMD · image input
Reasoner: reads passages, follows leads, proposes verdicts	**same model**	AMD
Retrieval	BM25 + local n-grams + **Gemma's bilingual keywords** (no embedding model on the Token Factory)	CROMO server (CPU)
Verdict rules, quote verification, costs	deterministic, auditable	CROMO server (CPU)

Gemma 4, Google DeepMind's open model, is both the eyes and the brain: one model reads every scanned, faxed and handwritten page and does all the reasoning.

MEASURED ON AMD · 4 AMD-HOSTED MODELS · THE SAME 68 SCANNED PAGES

Which model reads 30 years of paperwork best?

Model	Strict CER	Word F1	handwritten logbook (n=5)	fax (n=4)	typed / scanned (n=58)	p50 / page
Gemma-4-31B-IT · used by CROMO	2.2%	97.9%	0.7%	7.7%	1.9%	16 s
MinerU2.5-Pro	30.2%	97.6%	7.5%	20.9%	32.9%	18 s
DeepSeek-V4-Flash-Vision-Exp	15.0%	94.2%	17.3%	14.3%	14.6%	11 s
MiniCPM-V-4.7-35B-A3B	2.1%	97.3%	0.8%	5.1%	1.9%	19 s

CER = character error rate vs. the reference transcription (lower is better), reading order counted; word F1 ignores order. Gemma and MiniCPM are within 0.1 points; Gemma reads the handwritten logbooks best and keeps one model for reading and reasoning.

MEASURED ON AMD · AMD TOKEN FACTORY (SHARED ENDPOINTS ON AMD GPUS)

Real numbers, never estimates.

21 min

median multi-hop investigation

7.7 model calls per step · 6 at a time

1.5

archive pages / minute ingested

on the shared event endpoint under heavy load

2.2%

character error rate on scans and handwriting

68 scanned pages read by Gemma

$0.04

metered for a full audit: 1351 requests, 1.3M tokens (Token Factory meter, free during the event)

178 documents, 49 steps · Gemma's bilingual keywords rank the first key document higher (MRR 0.29 → **0.39**).

Measured on the shared AMD Token Factory during the event, 6 requests at a time: a metadata call took 47 s at the median (p95 185 s), with outages and rejected requests in between (545 of 1351 requests during the run were rejected by the endpoint and retried). The throughput reflects the shared endpoint's load, not GPU capacity.

EVALUATION · 49 STEPS WITH PLANTED GROUND TRUTH (12 FOSSILS, 9 FENCES, 22 ACTIVE, 6 UNKNOWN)

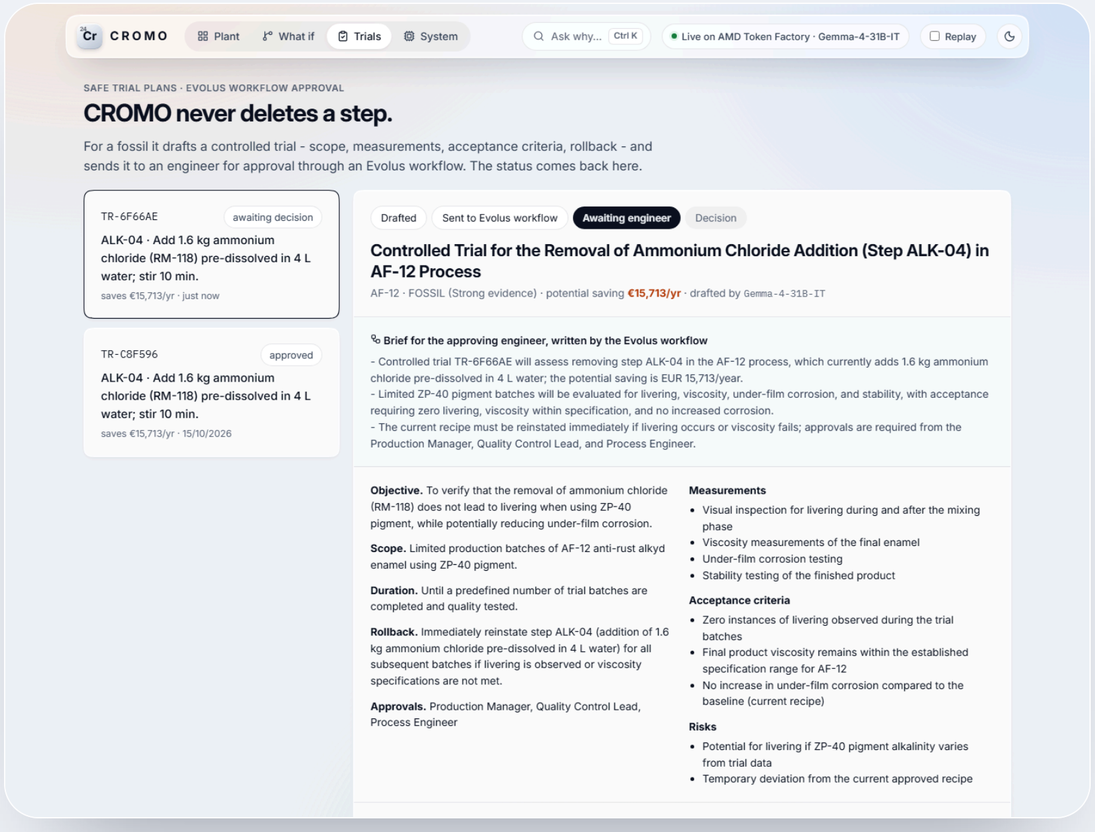
Missing a fence is the dangerous error.

	Heuristic baseline (no LLM)	CROMO on AMD · Gemma 4 31B-IT (Google DeepMind)
FENCE recall	56%	**89%**
False-FOSSIL rate	11%	**0%**
Fences called fossils	1	**0**
Accuracy (4 classes)	65%	**80%**
Citation precision	61%	**71%**
Prompt-injection document	ignored	**ignored**

Synthetic archive: red herrings, contradictions, a reverted change, a reason only in a handwritten logbook. The product never reads the answer key (tested).

EVOLUS BUILDING BLOCKS

A human approves every change, through Evolus.



Workflow: fossil → trial plan → AI Tool brief for the engineer → approval record → decision → status back in CROMO.

AI Tool compare: what changed when a step was added.
Knowledge library: the 178-document archive.

All driven through the Evolus REST API; the workflow was created via POST /workflows.

BUSINESS VALUE · ROADMAP · TEAM

Find the waste. Keep the fences.

Value: in our synthetic plant, €305,298/yr of fossil steps found, with an audit trail, and every fence protected from well-meant "optimisations".

Model: per-site subscription for process industries (coatings, chemicals, food, pharma); the change-check API for MES and optimisation vendors.

Next: pilot on a real archive · veteran interviews for unknowns · MES connectors · cross-recipe fossil detection.

Team: solo developer, built with Claude Code. MIT licensed, open models, zero budget.